LASER / MULTI-TRIGGER SECURITY SYSTEM

Curiosity Kickoff

- What happens when laser beam is not blocked?
- Can we see the laser?
- Where is it used?
-

Concept Connection

A Laser / Multi-Trigger Security System uses laser beams or multiple sensors to detect unauthorized access or interruptions. When the beam is broken or a trigger is activated, the system sends an alert through an alarm or LED, providing effective security and monitoring for protected areas.

1.Watt SMD LED

I'm flat and powerful. Just one of me can light up your whole table like a torch.I don't like wasting energy, so I shine super bright without getting too hot,You've already met me — I live in your mobile flashlight!

2.Buzzer

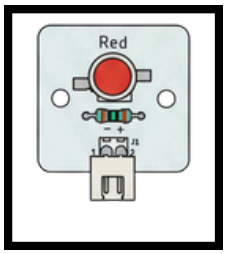
A sound-making device that works when electricity makes a thin metal plate vibrate rapidly inside. The vibration creates buzzing or beeping sounds. Some buzzers can even change frequency to play simple tunes or melodies.

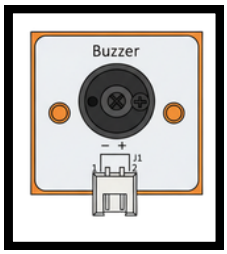
3.LDR Sensor :

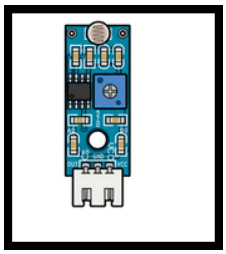
A light sensor whose resistance changes depending on light intensity. Lower resistance in bright light, higher in darkness.
Pinout: 4 Pins - VCC (+), Ground (-), Digital Pin (D0) & Analog Pin (A0)

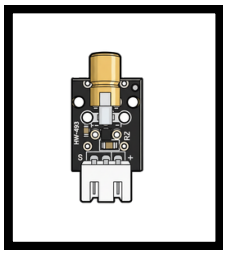
4. Laser Diode :

A small module that emits a focused red laser beam, often used in alignment, pointers, or security projects.Pinout: VCC – Power supply (+5V), GND – Ground, S (Signal/IN) – Control pin (to turn laser ON/OFF using a microcontroller)









Beginner: Laser Beam Alert

Class Explanation

A Laser Beam Alert system uses a laser beam and a sensor to detect interruptions. When the beam is broken by an object or person, the sensor triggers an alert such as an alarm or LED, making it useful for security and intrusion detection applications.

Materials Required

- Arduino with attached shield
- 1 × LDR sensor
- 1 × Lesar diode
- 1 × white 1 Watt SMD LED on breakout board
- RMC wires
- USB cable

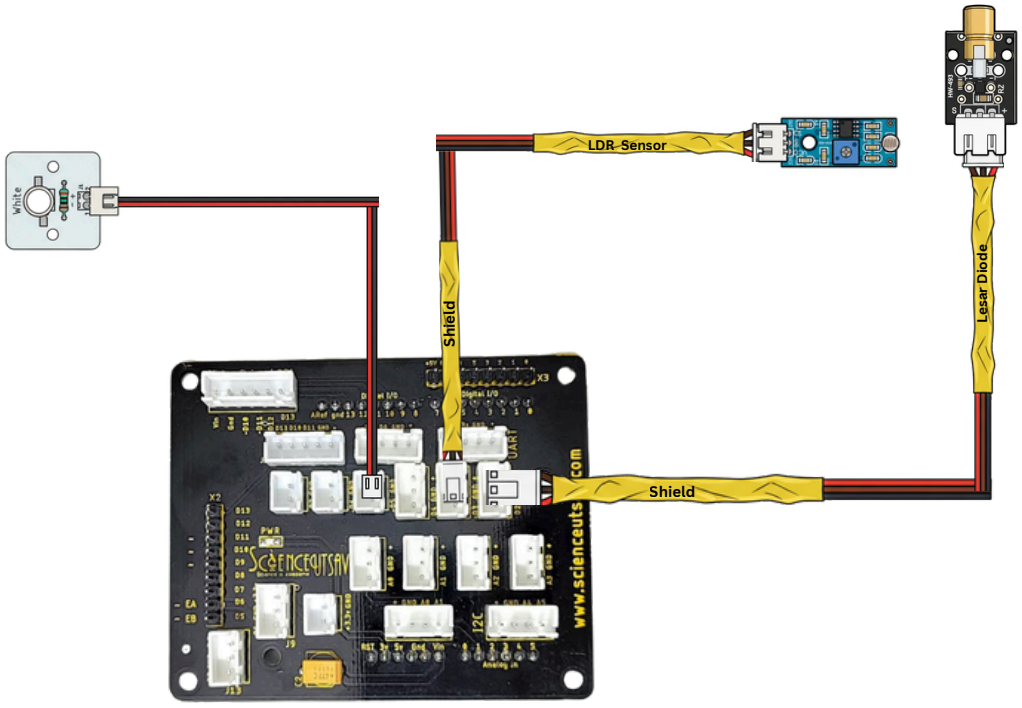
How to Connect

Use RMC wires to connect :

- 1 × LDR sensor→ Arduino pin D3 and GND
- 1 × white 1 Watt SMD → Arduino pin D5 and GND
- 1 × Lesar diode→ Arduino pin D2 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3
#define LED_PIN 5     // LED on D5

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON
  digitalWrite(LED_PIN, LOW);    // LED OFF initially
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == HIGH) {
    // Laser hitting LDR
    digitalWrite(LED_PIN, HIGH);  // LED ON
  } else {
    // Laser NOT hitting LDR
    digitalWrite(LED_PIN, LOW);   // LED OFF
  }
}
```

Code Explanation

- In this Arduino program, a laser diode is connected to pin 2 (LASER_PIN), an LDR sensor to pin 3 (LDR_PIN), and an LED to pin 5 (LED_PIN).
- The setup() function runs once when the Arduino starts. It sets the laser and LED as OUTPUT and the LDR as INPUT using pinMode(). The laser is turned ON, and the LED is initially OFF.
- The loop() function runs continuously. It reads the LDR value using digitalRead() and stores it in ldrState.
- If the laser beam is hitting the LDR (HIGH), the LED turns ON. If the beam is not detected (LOW), the LED turns OFF.
- So overall, the LED indicates whether the laser beam is reaching the LDR or not.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Laser Beam Alert Basics but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3
#define BUZZER_PIN 5  // Buzzer on D5

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON
  digitalWrite(BUZZER_PIN, LOW); // Buzzer OFF initially
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == HIGH) {
    // Laser hitting LDR
    digitalWrite(BUZZER_PIN, HIGH); // Buzzer ON
  } else {
    // Laser NOT hitting LDR
    digitalWrite(BUZZER_PIN, LOW);  // Buzzer OFF
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

the basic idea of Laser Beam Alert Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Intermediate: Laser Security Shield

Class Explanation

A Laser Security Shield system uses laser beams and sensors to create a protective barrier. When the beam is interrupted, it triggers an alert such as an alarm or LED, helping detect intrusions and enhance security in restricted or protected areas.

Materials Required

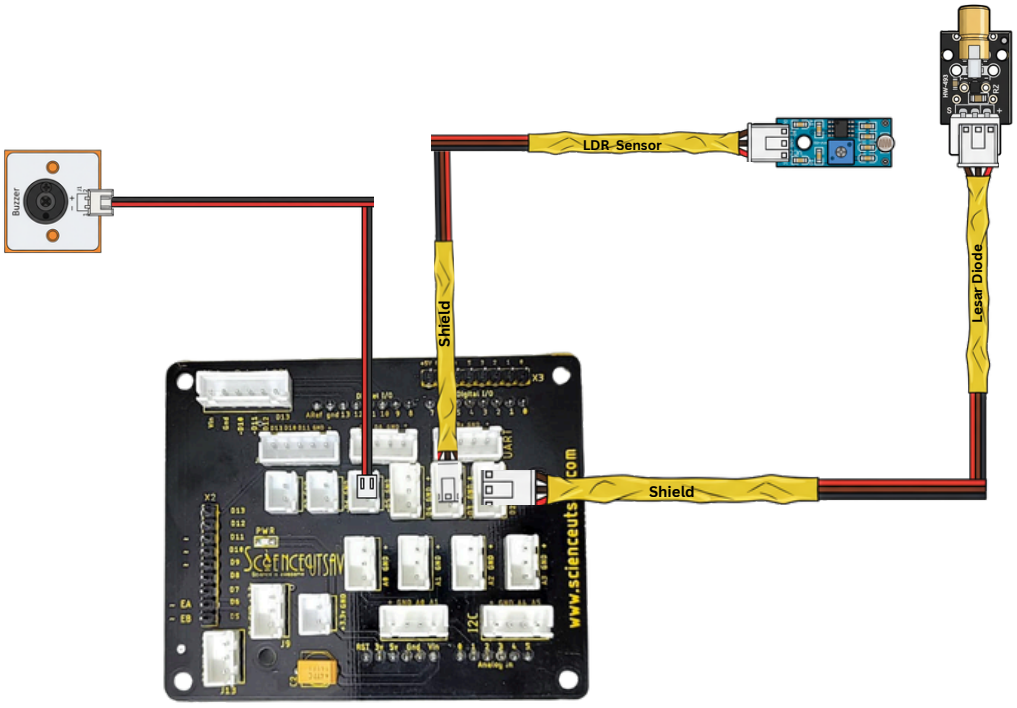
- Arduino with attached shield
- 1 × LDR sensor
- 1 × Lesar diode
- 1 × Buzzer
- RMC wires
- USB cable

How to Connect

- Use RMC wires to connect :
- 1 × LDR sensor→ Arduino pin D3 and GND
- 1 × Lesar diode→ Arduino pin D2 and GND
- 1 × Buzzer→ Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define LASER_PIN 2 // Laser diode connected to D2
#define LDR_PIN 3  // LDR sensor connected to D3
#define BUZZER_PIN 5 // Buzzer connected to D5

void setup() {
 pinMode(LASER_PIN, OUTPUT);
 pinMode(LDR_PIN, INPUT);
 pinMode(BUZZER_PIN, OUTPUT);
 digitalWrite(LASER_PIN, HIGH); // Turn ON laser initially
 digitalWrite(BUZZER_PIN, LOW); // Keep buzzer OFF
}

void loop() {
 int ldrState = digitalRead(LDR_PIN); // Read LDR state

 if (ldrState == HIGH) {       // Laser beam detected normally
  digitalWrite(BUZZER_PIN, LOW);   // Buzzer OFF
 } else {              // Laser beam blocked
  digitalWrite(BUZZER_PIN, HIGH);  // Buzzer ON (Alarm)
 }
}
```

Code Explanation

- In this Arduino program, a laser diode is connected to pin 2 (LASER_PIN), an LDR sensor is connected to pin 3 (LDR_PIN), and a buzzer is connected to pin 5 (BUZZER_PIN).
- The setup() function runs once when the Arduino starts. It sets the laser as OUTPUT, the LDR as INPUT, and the buzzer as OUTPUT using pinMode(). The laser is turned ON initially, and the buzzer is kept OFF.
- The loop() function runs continuously. It reads the LDR sensor value using digitalRead() and stores it in ldrState.
- If the laser beam is continuously detected by the LDR (HIGH), the buzzer remains OFF. If the beam is interrupted or blocked (LOW), the buzzer turns ON to indicate an alarm.

So overall, the system works like a security alarm that activates the buzzer when the laser beam is broken.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Laser Security Shield with Arduino but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3
#define LED_PIN 5     // LED on D5

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON
  digitalWrite(LED_PIN, LOW);    // LED OFF initially
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == HIGH) {
    digitalWrite(LED_PIN, LOW);   // Beam normal → LED OFF
  } else {
    digitalWrite(LED_PIN, HIGH);  // Beam broken → LED ON
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:
the basic idea of Laser Security Shield with Arduino how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Advanced: Smart Security Panel

Class Explanation

A Smart Security Panel is an integrated system that uses multiple sensors and controls to monitor and protect an area. It can detect events like motion, light changes, or intrusions and trigger alerts through alarms or displays, providing efficient and intelligent security management.

Materials Required

- Arduino with attached shield
- 1 × LDR sensor
- 1 × Lesar diode
- 1 × White 1 Watt SMD LED on breakout board
- 1 × Buzzer
- RMC wires
- USB cable

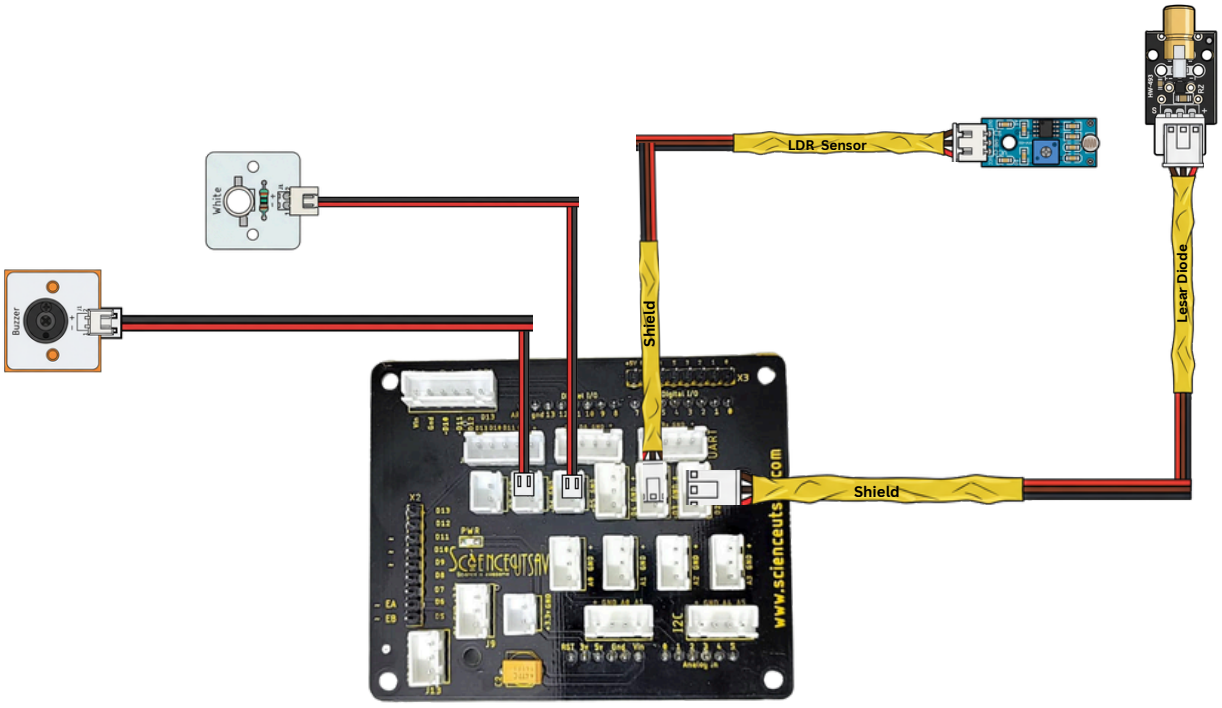
How to Connect

Use RMC wires to connect :

- 1 × LDR sensor → Arduino pin D3 and GND
- 1 × Lesar diode → Arduino pin D2 and GND
- 1 × Buzzer → Arduino pin D6 and GND
- 1 × White 1 Watt SMD LED → Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3
#define LED_PIN 5     // LED on D5
#define BUZZER_PIN 6  // Buzzer on D6

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON
  digitalWrite(LED_PIN, LOW);    // LED OFF initially
  digitalWrite(BUZZER_PIN, LOW); // Buzzer OFF initially
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == HIGH) {
    // Beam normal
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
  } else {
    // Beam broken → Alarm
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
  }
}
```

Code Explanation

- In this Arduino program, a laser diode is connected to pin 2 (LASER_PIN), an LDR sensor to pin 3 (LDR_PIN), an LED to pin 5 (LED_PIN), and a buzzer to pin 6 (BUZZER_PIN).

- The setup() function runs once when the Arduino starts. It sets the laser, LED, and buzzer as OUTPUT, and the LDR as INPUT using pinMode(). The laser is turned ON, while the LED and buzzer are initially OFF.

- The loop() function runs continuously. It reads the LDR value using digitalRead() and stores it in ldrState.

- If the laser beam is properly received (HIGH), both the LED and buzzer remain OFF. If the beam is broken (LOW), the LED and buzzer turn ON to indicate an alarm.

So overall, the system acts as a security alarm that activates both LED and buzzer when the laser beam is interrupted.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Smart Security Panel but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3

#define LED1_PIN 5    // LED1 on D5
#define LED2_PIN 6    // LED2 on D6

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);

  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON

  digitalWrite(LED1_PIN, LOW);
  digitalWrite(LED2_PIN, LOW);
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);

  if (ldrState == HIGH) {
    // Beam normal
    digitalWrite(LED1_PIN, LOW);
    digitalWrite(LED2_PIN, LOW);
  } else {
    // Beam broken
    digitalWrite(LED1_PIN, HIGH);
    digitalWrite(LED2_PIN, HIGH);
  }
}
```

Obective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:

the basic idea of Smart Security Panel how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Expert: Multi Sensor Fusion System

Class Explanation

A Multi Sensor Fusion System combines data from multiple sensors such as light, sound, motion, or magnetic sensors to make accurate decisions. It processes inputs together using logic control, improving reliability and enabling smart responses like alerts, automation, and advanced monitoring in electronic systems.

Materials Required

- Arduino with attached shield
- 1 × IR sensor
- 1 × LDR sensor
- 1 × Lesar Diode
- 2 × White 1 Watt SMD LED on breakout board
- RMC wires
- USB cable

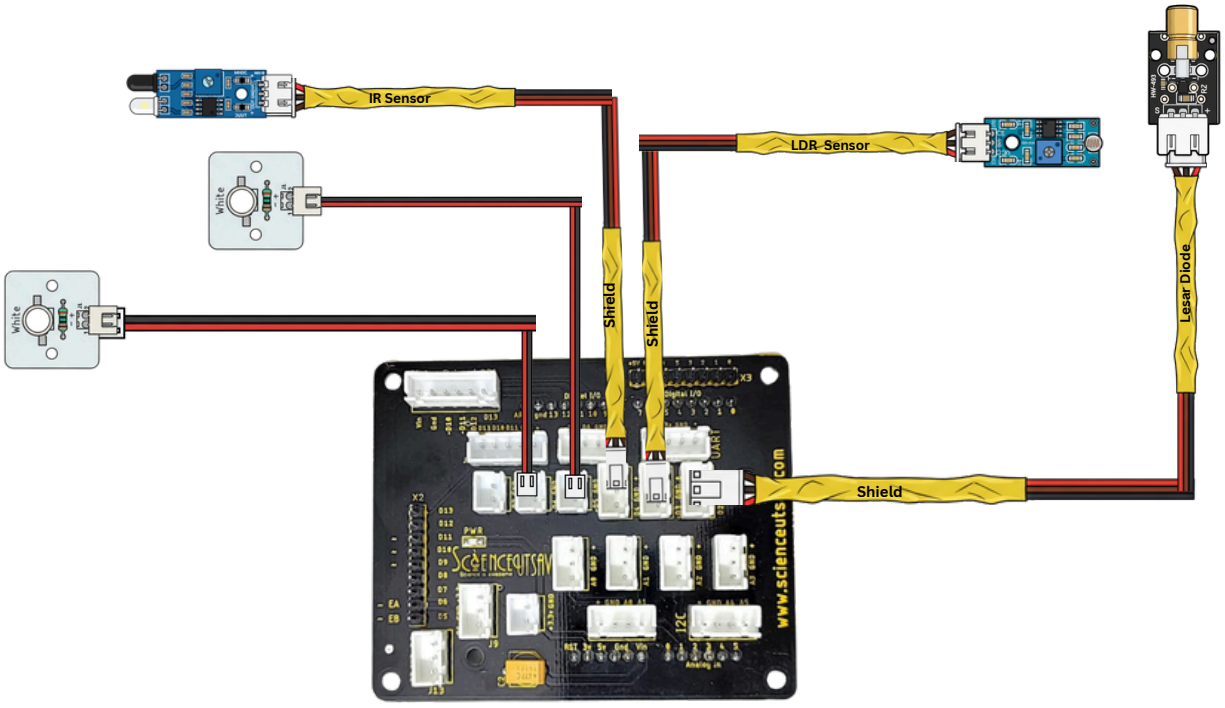
How to Connect

Use RMC wires to connect :

- 1 × IR sensor → Arduino pin D4 and GND
- 1 × LDR sensor → Arduino pin D3 and GND
- 1 × Lesar Diode→ Arduino pin D2 and GND
- 1 × White 1 Watt SMD LED → Arduino pin D5 and GND
- 1 × White 1 Watt SMD LED → Arduino pin D6 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3
#define IR_PIN 4      // IR sensor on D4

#define LED1_PIN 5    // LED1 on D5
#define LED2_PIN 6    // LED2 on D6

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(IR_PIN, INPUT);

  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON

  digitalWrite(LED1_PIN, LOW);
  digitalWrite(LED2_PIN, LOW);
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);
  int irState = digitalRead(IR_PIN);

  // IR usually LOW when object detected
  bool irDetected = (irState == LOW);
  bool beamBroken = (ldrState == LOW);

  if (beamBroken || irDetected) {
    // Alarm condition
    digitalWrite(LED1_PIN, HIGH);
    digitalWrite(LED2_PIN, HIGH);
  } else {
    // Normal condition
    digitalWrite(LED1_PIN, LOW);
    digitalWrite(LED2_PIN, LOW);
  }
}
```

Code Explanation

- In this Arduino program, a laser diode is connected to pin 2 (LASER_PIN), an LDR sensor to pin 3 (LDR_PIN), and an IR sensor to pin 4 (IR_PIN). Two LEDs are connected to pins 5 (LED1_PIN) and 6 (LED2_PIN).

- The setup() function runs once when the Arduino starts. It sets the laser and LEDs as OUTPUT, and both sensors (LDR and IR) as INPUT using pinMode(). The laser is turned ON, and both LEDs are initially OFF.

- The loop() function runs continuously. It reads the LDR and IR sensor values using digitalRead().

- If the laser beam is broken (LDR LOW) or an object is detected by the IR sensor (LOW), it is considered an alarm condition. In this case, both LEDs turn ON. Otherwise, both LEDs remain OFF.

So overall, the system uses two sensors for detection, and turns ON both LEDs if either the laser beam is interrupted or an object is detected.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Multi Sensor Fusion System but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define LASER_PIN 2   // Laser diode on D2
#define LDR_PIN 3     // LDR sensor on D3
#define IR_PIN 4      // IR sensor on D4

#define BUZZER_PIN 5  // Buzzer on D5

void setup() {
  pinMode(LASER_PIN, OUTPUT);
  pinMode(LDR_PIN, INPUT);
  pinMode(IR_PIN, INPUT);

  pinMode(BUZZER_PIN, OUTPUT);

  digitalWrite(LASER_PIN, HIGH); // Laser ON
  digitalWrite(BUZZER_PIN, LOW); // Buzzer OFF
}

void loop() {
  int ldrState = digitalRead(LDR_PIN);
  int irState = digitalRead(IR_PIN);

  bool irDetected = (irState == LOW);   // IR detection
  bool beamBroken = (ldrState == LOW);  // Laser interruption

  if (beamBroken || irDetected) {
    digitalWrite(BUZZER_PIN, HIGH); // Alarm ON
  } else {
    digitalWrite(BUZZER_PIN, LOW);  // Alarm OFF
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn:
the basic idea of Multi Sensor Fusion System how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Learning Outcomes
After completing this session, students should be able to:

- Understand how laser beams and sensors detect interruptions and trigger alerts for basic security and intrusion detection systems.
- Learn how multiple laser beams create a protective barrier and activate alarms when the beam is broken.
- Understand how different sensors are integrated to monitor conditions and provide alerts for improved security management.
- Learn how data from multiple sensors is combined and processed to improve accuracy and enable intelligent decision-making in electronic systems.

Try Out Other Things

- use the Try This Yourself code after each main class code before asking students to build a completely new variation
- ask students to point out the exact lines that changed between the first code and the Try This Yourself code
- let students first run the Try This Yourself code code as it is and then change one value, one condition, or one output